

EV BATTERIES

CHARGED UP

The prices of EV batteries are falling faster than what was anticipated earlier. This could result in more affordable battery packs

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► **INDIA EMBARKED ON** its electric vehicles (EV) journey more than two decades ago. However, the trend gained momentum only after Tata Motors plugged into the market with its Nexon.ev in December 2019. Since then, sales of four-wheeler EVs have been averaging around 7,000 units a month. And sales of electric two-wheelers have been comfortably hitting the 75,000-mark each month.

Most of the EVs sold over the past four years came with battery guarantees. Tata Motors and Ola Electric, both industry leaders, offer replacement warranties of eight years or 80,000 km, whichever occurs first. For many customers, EV batteries will become depleted in the next few years, and have to be changed. Replacing a battery will be an expensive affair as a considerable part of the overall cost of an EV is the cell. For instance, a battery pack for the Nexon.ev—beyond the warranty period—costs around ₹7 lakh, reports say.

The good news is that battery prices will fall—faster than what was anticipated earlier.

A research report from lithium-ion cell intelligence firm Benchmark Minerals pegs the cost of batteries at \$78 per kilowatt hour (kWh) as of September 2024. The cost of a 30-kWh battery pack for a Nexon.ev would be \$2,340 or ₹1,96,560, based on a conversion rate of ₹84 a dollar. Four years ago, the same battery would have cost ₹2,77,200, as the price of lithium-ion batteries

